ONLINE-ONLY SUPPLEMENTS

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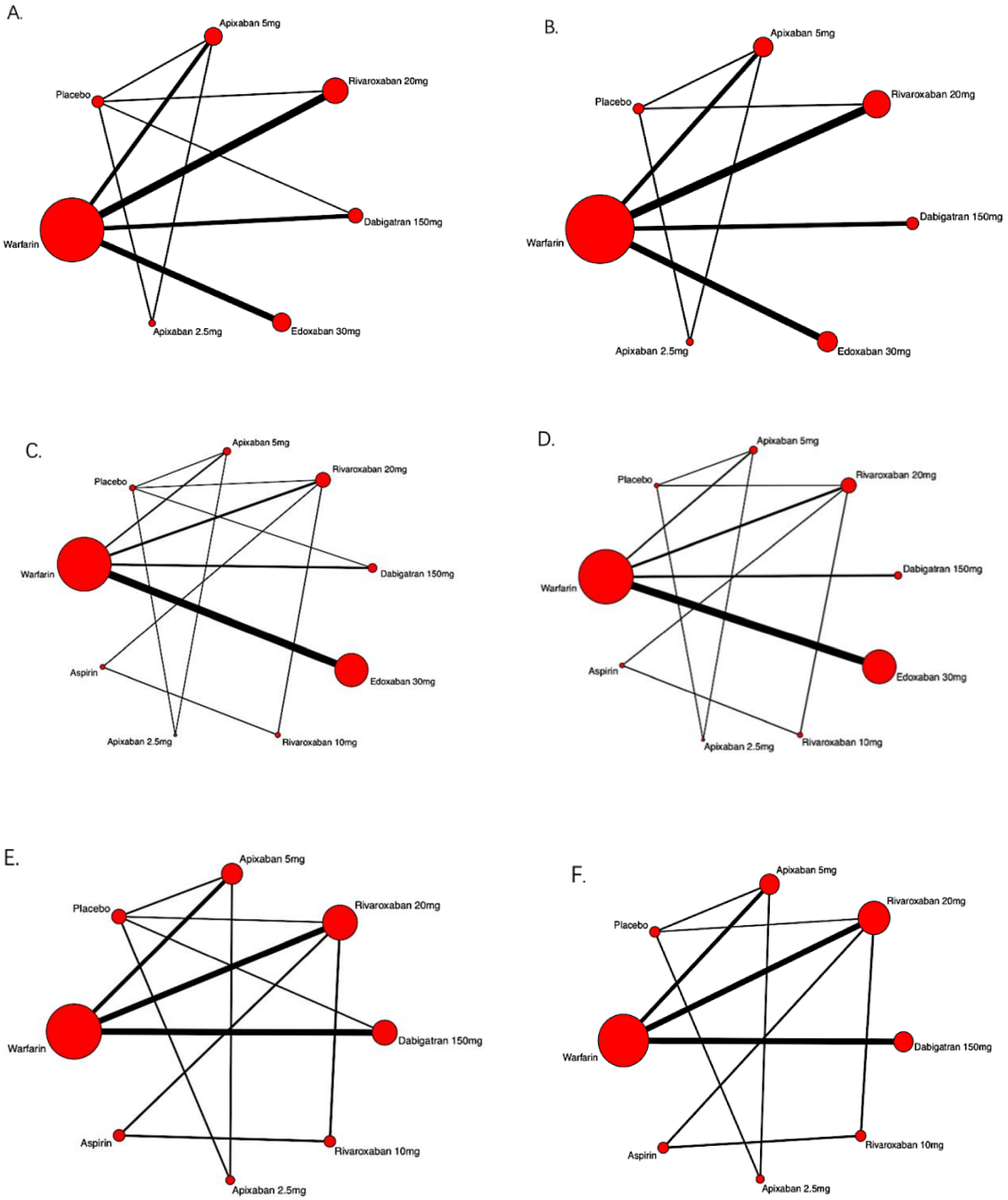
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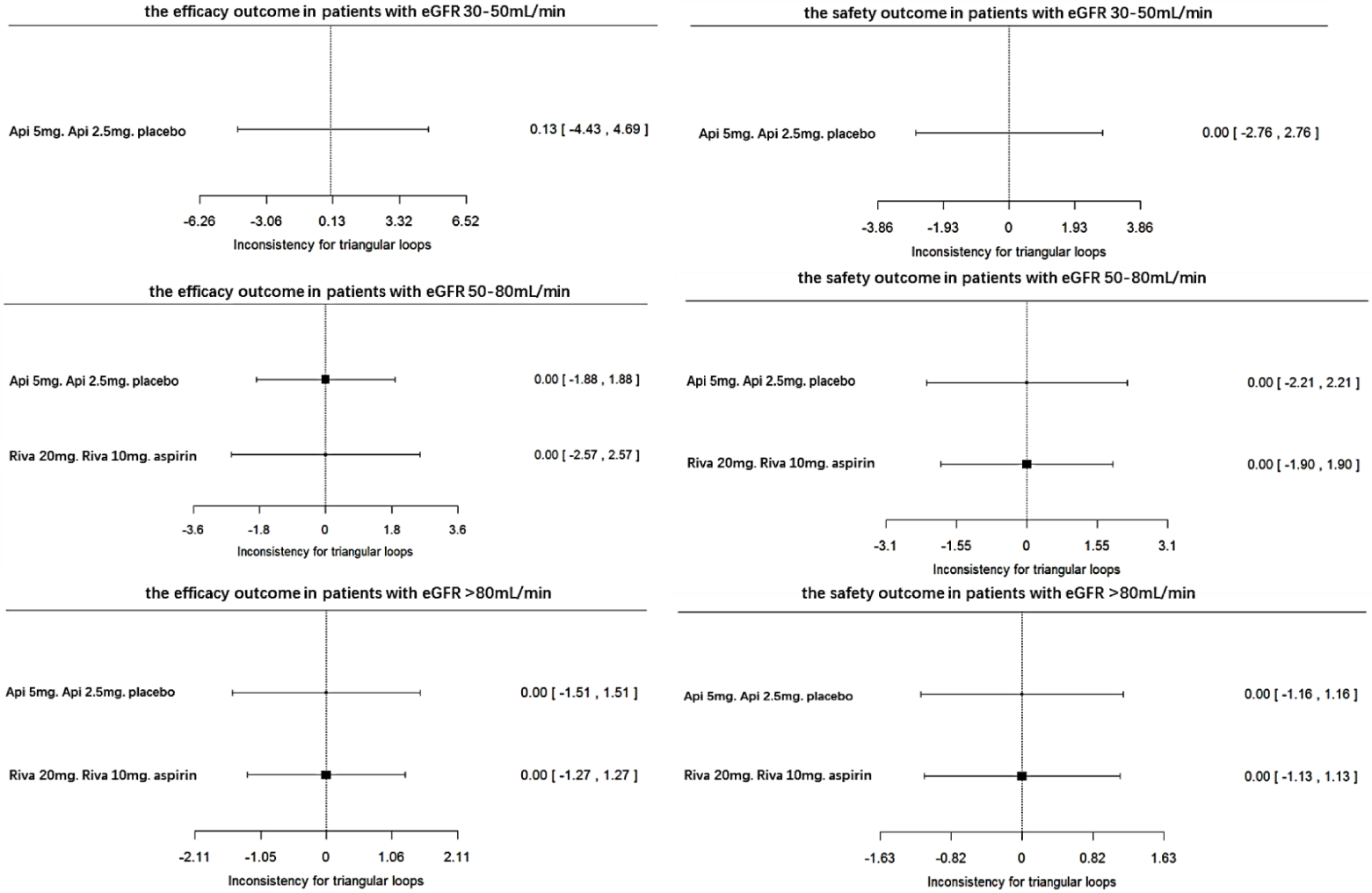
Supplemental Item 1. Search Strategy



Supplemental Figure 1. Network of treatment comparisons for Bayesian network meta-analysis.

A. The network of the primary efficacy outcome in VTE patients with CrCl 30-50 mL/min; B. The network of the primary safety outcome in VTE patients with CrCl 30-50 mL/min; C. The network of the primary efficacy outcome in VTE patients with CrCl 50-80 mL/min; D. The network of the safety outcome in VTE patients with CrCl 50-80 mL/min; E. The network of the primary efficacy outcome in VTE patients with CrCl more than 80 mL/min; F. The network of the safety outcome in VTE patients with CrCl more than 80 mL/min. Directly comparable treatments are linked with a line, the width of which is proportional to the number of trials comparing the connected treatments. Apixaban and dabigatran were administered twice daily, while the other treatments were administered once daily. CrCl = creatinine clearance; VTE = venous thromboembolism.

Supplemental Figure 2. Assessment of inconsistency



We estimated inconsistency as the difference between direct and indirect estimates (called inconsistency factor, IF) and the corresponding 95% confidence intervals (CI) for IF in each closed loop. The following graphs show all closed triangular loops (loops formed by three treatments) in efficacy and safety outcome network. Inconsistent loops are those that present IF with 95% CIs incompatible with zero. There are no inconsistent loops in all outcomes.

Supplemental Table 1. Definitions of the outcomes in included trials

Study	the Efficacy Outcome	the Safety Outcome
Hokusai-VTE	The primary efficacy outcome was **the incidence of adjudicated symptomatic recurrent venous thromboembolism,** which was defined as a composite of deep-vein thrombosis or nonfatal or fatal pulmonary embolism.	The pre-specified outcome measures for bleeding were **the composite of major and clinically relevant non-major bleeding** during the study treatment period**. Major bleeding** was defined as acute clinically overt bleeding accompanied by one or more of the following: a decrease in the blood hemoglobin level of 2 g/dL or more; transfusion of two or more units of packed red blood cells; critical site bleeding; bleeding into the operated joint requiring reoperation or intervention; intramuscular bleeding with compartment syndrome; or fatal bleeding**. Clinically relevant non-major bleeding** did not meet criteria for major bleeding and met the descriptions provided, including epistaxis, gastrointestinal bleed, hematuria, bruising/ecchymosis, hemoptysis, and hematoma**.**
AMPLIFY	The primary efficacy outcome was the incidence of the **adjudicated composite of recurrent symptomatic venous thromboembolism or death related to venous thromboembolism.** Recurrent venous thromboembolism included fatal or non- fatal pulmonary embolism and deep-vein thrombosis**.** Death was adjudicated as related to venous thromboembolism, related to cardiovascular disease, caused by bleeding, or due to other causes.	The primary safety outcome was **adjudicated major bleeding.** Bleeding was defined as major if it was overt and associated with a decrease in the hemoglobin level of 2 g per deciliter or more, required the transfusion of 2 or more units of blood, occurred into a critical site, or contributed to death.
AMPLIFY-EXT		**A composite of major bleeding and clinically relevant nonmajor bleeding. Major leeding** was defined as a decrease in the hemoglobin level of 2 g per deciliter or more, required the transfusion of 2 or more units of blood, occurred into a critical site, or contributed to death**. Clinically relevant nonmajor bleeding** was defined as overt bleeding that did not meet the criteria for major bleeding but that was associated with the need for medical intervention, unscheduled contact with a physician, interruption or discontinuation of the study drug, or discomfort or impairment of activities of daily living**.**
RE-MEDY	The primary efficacy outcome was **recurrent symptomatic and objectively verified venous thromboembolism or death associated with venous thromboembolism.** Clinically suspected recurrent deep-vein thrombosis had to be objectively verified using pre-specified imaging studies.	Safety outcomes included **major bleeding and clinically relevant nonmajor bleeding. Major bleeding** was defined as clinically overt and associated with a fall of the hemoglobin level of 20 g/L or required transfusion of at least 2 units of red cells or, involved a critical organ or was fatal**. Clinically relevant non-major bleeding**: At least one of the following criteria had to be fulfilled: 1) Spontaneous skin hematoma of at least 25 cm; 2) Spontaneous nose bleed of more than 5 minutes duration; 3) Macroscopic hematuria, either spontaneous or, if associated with an intervention; 4) Spontaneous rectal bleeding; 5) Gingival bleeding for more than 5 minutes; 6)Bleeding leading to hospitalization and/or requiring surgical treatment; 7) Bleeding leading to a transfusion of less than 2 units of whole blood or red cells; 8) Any other bleeding event considered clinically relevant by the investigator
RE-SONATE		
EINSTEIN-EXT/ EINSTEIN-DVT	The primary efficacy outcome was **symptomatic, recurrent venous thromboembolism**, defined as the composite of deep-vein thrombosisor nonfatal or fatal pulmonary embolism**. Recurrent pulmonary embolism** were defined as one or more of the following findings: a new intraluminal filling defect on spiral CT or pulmonary angiography, a cutoff of a vessel of more than 2.5 mm in diameter on pulmonary angiography, a new perfusion defect of at least 75% of a segment with corresponding normal ventilation, a new non–high-probability perfusion defect associated with deep venous thrombosis as documented by ultrasonography or venography, or a new pulmonary embolism confirmed at autopsy. **Recurrent deep venous thrombosis** was defined as one or more of the following findings: a new noncompressible venous segment or a substantial increase (4 mm or more) in the diameter of the thrombus during full compression in a previously abnormal segment on ultrasonography or a new intraluminal filling defect on venography**.**	**The composite of major or clinically relevant nonmajor bleeding. Major bleeding:** if it was clinically overt and was associated with a decrease in the hemoglobin level of ⩾2.0 g/dl; if bleeding led to the transfusion of ⩾2 units of red cells; or if bleeding was intracranial or retroperitoneal, occurred in another critical site, or contributed to death**. Clinically relevant nonmajor bleeding** was defined as overt bleeding that did not meet the criteria for major bleeding but was associated with medical intervention, unscheduled contact with a physician, interruption or discontinuation of a study drug, or discomfort or impairment of daily activities**.**
RE-COVER I/II	The primary efficacy outcome was **recurrent, symptomatic objectively confirmed venous thromboembolism (i.e., pulmonary embolism or deep-vein thrombosis) or venous thromboembolism related death**, assessed from randomization up to the end of treatment.	A composite of **major bleeding and clinically relevant nonmajor bleeding. Major bleeding** was defined according to International Society on Thrombosis and Haemostasis (ISTH) criteria**.** Definition of **clinically relevant non-major bleeding**: At least one of the following criteria had to be fulfilled: 1) Spontaneous skin hematoma of at least 25 cm2; 2) Spontaneous nose bleed of more than 5 minutes duration; 3) Macroscopic hematuria, either spontaneous or, if associated with an intervention, lasting more than 24 hours; 4) Spontaneous rectal bleeding (more than spotting on toilet paper); 5) Gingival bleeding for more than 5 minutes; 6) Bleeding leading to hospitalization and/or requiring surgical treatment; 7) Bleeding leading to a transfusion of less than 2 units of whole blood or red cells; 8) Any other bleeding event considered clinically relevant by the investigator**.**
EINSTEIN CHOICE/ EINSTEIN-PE	The primary efficacy outcome was **a composite of symptomatic, recurrent fatal or nonfatal venous thromboembolism and unexplained death for which pulmonary embolism could not be ruled out.** Recurrent venous thromboembolism included fatal and nonfatal pulmonary embolism and deep-vein thrombosis.	The principal safety outcome was **a composite of major or clinically relevant nonmajor bleeding. Major bleeding** was defined as overt bleeding that was associated with a decrease in the hemoglobin level of 2 g per deciliter or more, led to transfusion of 2 or more units of red cells, occurred in a critical site, or contributed to death**. Clinically relevant nonmajor bleeding** was defined as overt bleeding that did not meet the criteria for major bleeding but was associated with the need for medical intervention, unscheduled contact with a physician, interruption or discontinuation of the study drug, or discomfort or impairment of activities of daily living

Supplemental Table 2. Characteristics of included trials

Study	Year	Age	Male (%)	Sample size	Anticoagulant indication	Follow-up (months)	Excluded CrCl profile (mL/min)	Subgroup of CrCl (mL/min)	Intervention	Control	Efficacy outcomes	Safety outcomes
Hokusai-VTE	2013	56	57	8240	Acute symptomatic DVT and/or PE	12	<30	30-50 (n=541) >50 (n=7699)	Edoxaban 60mg once daily or 30mg once daily in patients with eGFR 30-50mL/min	Warfarin, INR 2.0-3.0	Composite of DVT or PE	Major or clinically relevant minor bleeding
AMPLIFY	2013	57	59	4783	Symptomatic DVT or PE	6	<25	25-50 (n=327) 50-80 (n=1061) >80 (n=3395)	Apixaban 10mg twice daily for the first 7 days, followed by 5 mg twice daily for 6 months	Warfarin, INR 2.0-3.0	Recurrent VTE or death related to VTE	Major bleeding
AMPLIFY-EXT	2013	57	58	2413	Symptomatic DVT or PE treated for 6 to 12 months with anticoagulant therapy	6	<25	25-50 (n=138) 50-80 (n=536) >80 (n=1739)	Apaixaban 2.5 or 5mg twice daily	Placebo	Recurrent VTE or death related to VTE	Major bleeding
RE-MEDY	2013	59	61	2824	Symptomatic DVT/PE already treated with anticoagulant or treated in RE-COVER I/II trials	6	<30	30-50 (n=104) 50-80 (n=617) >80 (n=2103)	Dabigatran 150 mg twice daily	Warfarin, INR 2.0-3.0	Recurrent VTE or death related to VTE	None
RE-SONATE	2013	56	55	1339	Symptomatic DVT/PE already treated with anticoagulant or treated in RE-COVER I/II trials	18	<30	30-50 (n=71) 50-80 (n=334) >80 (n=934)	Dabigatran 150 mg twice daily	Placebo	Recurrent VTE or death related to VTE	None
EINSTEIN-DVT	2010	56	56	3405	Acute symptomatic DVT	12	<30	30-50 (n=250) 50-80 (n=792) >80 (n=2363)	Rivaroxaban 15 mg twice daily for 3 weeks then 20 mg once daily	Warfarin, INR 2.0-3.0	DVT or non-fatal or fatal PE	Major or clinically relevant minor bleeding
EINSTEIN-EXT	2010	58	58	1088	Symptomatic DVT or PE who had been treated for 6 or 12 months with a vitamin K antagonist or rivaroxaban	12	<30	30-50 (n=86) 50-80 (n=256) >80 (n=746)	Rivaroxaban 20 mg once daily	Placebo	DVT or non-fatal or fatal PE	Major or clinically relevant minor bleeding
EINSTEIN-PE	2012	58	53	4806	Acute symptomatic PE	12	<30	30-50 (n=404) 50-80 (n=1230) >80 (n=3172)	Rivaroxaban 15 mg twice daily for 3 weeks then 20 mg once daily	Warfarin, INR 2.0-3.0	Recurrent VTE	Major or clinically relevant minor bleeding
RE-COVER I/II	2017	55	60	5035	Acute symptomatic proximal DVT or PE	6	<30	30-50 (n=237) 50-80 (n=1100) >80 (n=3698)	Dabigatran 150 mg twice daily	Warfarin, INR 2.0-3.0	Recurrent VTE or death related to VTE	Major or clinically relevant minor bleeding
EINSTEIN CHOICE	2017	58	55	3365	Symptomatic DVT or PE treated for 6 to 12 months with anticoagulant therapy	12	<30	30-50 (n=156) 50-80 (n=858) >80 (n=2351)	Rivaroxaban 10mg or 20mg once daily	Aspirin	Recurrent VTE	Major or clinically relevant minor bleeding

CKD = chronic kidney disease; CrCl = creatinine clearance; DVT = deep vein thrombosis; INR = international normalized ratio; PE = pulmonary embolism; VTE = venous thromboembolism.

Supplemental Table 3. Assessment of risk of bias in included trials

Component Study	Bias arising from the randomization process	Bias due to deviations from intended interventions	Bias due to missing outcome data	Bias in measurement of the outcome	Bias in selection of the reported result	Risk-of-bias judgment
Hokusai-VTE	Low	Low	Low	Low	Low	Low
AMPLIFY	Low	Low	Low	Low	Low	Low
AMPLIFY-EXT	Low	Low	Low	Low	Low	Low
RE-MEDY	Low	Low	Low	Low	Low	Low
RE-SONATE	Low	Low	Low	Low	Low	Low
EINSTEIN-DVT	Low	Low	High	Low	Low	Some concern
EINSTEIN-EXT	Low	Low	Low	Low	Low	Low
EINSTEIN-PE	Low	Low	High	Low	Low	Some concern
RE-COVER I/II	Low	Low	Low	Low	Low	Low
EINSTEIN CHOICE	Low	Low	Low	Low	Low	Low

Supplemental Table 4. Summary strength of evidence (SOE) ratings for directed comparisons

Quality assessment								No of patients		Effect	Quality
Populations	**Outcomes**	**No of studies**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**Intervention**	**Control**	**OR (95% CI)**	
the treatment of acute VTE	VTE or VTE-related death	5	no serious risk of bias	no serious inconsistency	no serious indirectness	no serious imprecision	none	341/13117	362/13152	0.96 (0.82 to 1.11)	⊕⊕⊕⊕
								2.6%	2.8%		HIGH
	Bleeding events	5	no serious risk of bias	serious	no serious indirectness	no serious imprecision	none	1099/13075	1374/13107	0.76 (0.68 to 0.90)	⊕⊕⊕O
								8.4%	10.4%		MODERATE
the extended treatment of VTE	VTE or VTE-related death	4	no serious risk of bias	no serious inconsistency	no serious indirectness	serious	none	44/3581	125/2189	0.23 (0.16 to 0.29)	⊕⊕⊕O
								1.2%	5.7%		MODERATE
	Bleeding events	3	no serious risk of bias	serious	no serious indirectness	serious	none	159/4382	50/2744	1.86 (1.04 to 3.30)	⊕⊕O
								3.6%	1.8%		LOW

Abbreviations: CI = confidence intervals; OR = odds ratio; VTE = venous thromboembolism.

Population	Outcomes	Model assumption	*Dbar*	*Pd*	*DIC*	Accept Model
CrCl: 30-50 mL/min	the efficacy outcome	**Fixed consistency**	**76**	**15**	**91**	**Fixed consistency**
		Fixed inconsistency	78	16	94	
		Random consistency	75	17	92	
		Random inconsistency	78	17	95	
	the safety outcome	**Fixed consistency**	**83**	**14**	**97**	**Fixed consistency**
		Fixed inconsistency	84	15	99	
		Random consistency	83	15	98	
		Random inconsistency	83	16	99	
CrCl: 50-80 mL/min	the efficacy outcome	**Fixed consistency**	**113**	**18**	**131**	**Fixed consistency**
		Fixed inconsistency	118	18	136	
		Random consistency	110	22	132	
		Random inconsistency	112	22	134	
	the safety outcome	**Fixed consistency**	**112**	**17**	**129**	**Fixed consistency**
		Fixed inconsistency	113	17	130	
		Random consistency	112	19	131	
		Random inconsistency	113	20	133	
CrCl: >80 mL/min	the efficacy outcome	**Fixed consistency**	**117**	**16**	**133**	**Fixed consistency**
		Fixed inconsistency	118	16	134	
		Random consistency	115	18	133	
		Random inconsistency	117	19	136	
	the safety outcome	**Fixed consistency**	**108**	**16**	**124**	**Fixed consistency**
		Fixed inconsistency	109	16	125	
		Random consistency	109	16	125	
		Random inconsistency	111	16	127	

Supplemental Table 5. Evaluation of the model fit.

The table shows the mean posterior deviance (Dbar), the leverage (Pd, also termed the effective number of parameters) and the Deviance Information Criterion (DIC) of each model for outcome. The DIC means a measure of model fit that penalises model complexity – lower values of the DIC suggest a more parsimonious model.

Drugs	Efficacy			Safety		
	SUCRA	95%CrI	Rank	SUCRA	95%CrI	Rank
CrCl 30-50mL/min						
Dabigatran 150mg	0.88	0.33 to 1	1	0.51	0 to 1	3
Edoxaban 30mg	0.79	0.33 to 1	2	0.48	0 to 1	5
Apixaban 5mg	0.61	0.17 to 1	3	0.49	0 to 1	4
Rivaroxaban 20mg	0.47	0.17 to 0.83	4	0.38	0 to 0.83	6
Warfarin	0.43	0.17 to 0.67	5	0.13	0 to 0.5	7
Apixaban 2.5mg	0.29	0 to 1	6	0.67	0 to 1	2
Placebo	0.04	0 to 0.17	7	0.84	0 to 1	1
CrCl 50-80mL/min						
Rivaroxaban 20mg	0.85	0.5 to 1	1	0.294	0 to 0.63	8
Edoxaban 60mg	0.78	0.38 to 1	2	0.342	0.13 to 0.75	7
Warfarin	0.7	0.38 to 1	3	0.083	0 to 0.38	9
Dabigatran 150mg	0.53	0.13 to 1	4	0.446	0.13 to 0.75	6
Apixaban 5mg	0.51	0.13 to 1	5	0.67	0.13 to 0.88	2
Rivaroxaban 10mg	0.5	0.12 to 1	6	0.531	0 to 1	4
Apixaban 2.5mg	0.46	0.12 to 1	7	0.665	0 to 1	3
Aspirin	0.14	0 to 0.5	8	0.486	0 to 0.88	5
Placebo	0.05	0 to 0.25	9	0.983	0.87 to 1	1
CrCl >80mL/min						
Rivaroxaban 10mg	0.85	0.29 to 1	1	0.26	0 to 0.57	6
Rivaroxaban 20mg	0.64	0.29 to 1	2	0.11	0 to 0.29	7
Warfarin	0.61	0.29 to 1	3	0.1	0 to 0.29	8
Apixaban 5mg	0.56	0.14 to 1	4	0.75	0.71 to 1	3
Dabigatran 150mg	0.56	0.14 to 1	5	0.43	0.29 to 0.57	5
Apixaban 2.5mg	0.56	0.14 to 1	6	0.91	0.71 to 1	1
Aspirin	0.22	0 to 0.71	7	0.53	0.29 to 0.71	4
Placebo	0.01	0 to 0.14	8	0.9	0.71 to 1	2

Supplement Table 6. SUCRA and its 95% CrI for the primary efficacy and safety outcomes in VTE population with different CrCl.

Apixaban and Dabigatran were administrated twice daily, and others once daily. CrCl = creatinine clearance; CrI = credible intervals; SUCRA = surface under the cumulative ranking curve.

Supplement Table 7. Summary of confidence ratings for mixed and indirected comparisons in network meta-analysis.

Comparison	Number of studies	Within-study bias	Across-studies bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Mixed evidence of the efficacy outcome in patients with CrCl 30-50mL/min								
apixaban 2.5mg:apixaban 5mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 30mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 20mg	1	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
rivaroxaban 20mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Indirect evidence of the efficacy outcome in patients with CrCl 30-50mL/min								
apixaban 2.5mg:dabigatran 150mg	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:edoxaban 30mg	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
apixaban 5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:edoxaban 30mg	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
apixaban 5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:edoxaban 30mg	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 30mg:placebo	0	No concerns	Suspected	Some concerns	No concerns	Major concerns	No concerns	Very low
edoxaban 30mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
Mixed evidence of the efficacy outcome in patients with CrCl 50-80mL/min								
apixaban 2.5mg:apixaban 5mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:placebo	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 5mg:placebo	1	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
apixaban 5mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:rivaroxaban 10mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
dabigatran 150mg:placebo	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
dabigatran 150mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 60mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
rivaroxaban 10mg:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
rivaroxaban 20mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low

Comparison	Number of studies	Within-study bias	Across-studies bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Indirect evidence of the efficacy outcome in patients with CrCl 50-80mL/min								
apixaban 2.5mg:aspirin	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:edoxaban 60mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:warfarin	0	No concerns	Suspected	No concerns	Major concerns	No concerns	No concerns	Very low
apixaban 5mg:aspirin	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:edoxaban 60mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:dabigatran 150mg	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
aspirin:edoxaban 60mg	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
aspirin:placebo	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:edoxaban 60mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:rivaroxaban 10mg	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 60mg:placebo	0	No concerns	Suspected	Some concerns	No concerns	No concerns	No concerns	Very low
edoxaban 60mg:rivaroxaban 10mg	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
edoxaban 60mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:warfarin	0	No concerns	Suspected	Some concerns	No concerns	No concerns	No concerns	Very low
rivaroxaban 10mg:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
Mixed evidence of the efficacy outcome in patients with CrCl >80mL/min								
apixaban 2.5mg:apixaban 5mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:placebo	1	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
apixaban 5mg:placebo	1	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
apixaban 5mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:rivaroxaban 10mg	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
aspirin:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
dabigatran 150mg:placebo	1	No concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Low
dabigatran 150mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
rivaroxaban 10mg:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
rivaroxaban 20mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low

Comparison	Number of studies	Within-study bias	Across-studies bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Indirect evidence of the efficacy outcome in patients with CrCl >80mL/min								
apixaban 2.5mg:aspirin	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
apixaban 5mg:aspirin	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:placebo	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
aspirin:warfarin	0	No concerns	Suspected	Some concerns	No concerns	Major concerns	No concerns	Very low
dabigatran 150mg:rivaroxaban 10mg	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
placebo:warfarin	0	No concerns	Suspected	Some concerns	No concerns	No concerns	No concerns	Very low
rivaroxaban 10mg:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
Mixed evidence of the safety outcome in patients with CrCl 30-50mL/min								
apixaban 2.5mg:apixaban 5mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 30mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
rivaroxaban 20mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Indirect evidence of the safety outcome in patients with CrCl 30-50mL/min								
apixaban 2.5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:edoxaban 30mg	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
apixaban 2.5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:warfarin	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
apixaban 5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:edoxaban 30mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:edoxaban 30mg	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:placebo	0	No concerns	Suspected	No concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 30mg:placebo	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low
edoxaban 30mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low

Comparison	Number of studies	Within-study bias	Across-studies bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Mixed evidence of the efficacy outcome in patients with CrCl 50-80mL/min								
apixaban 2.5mg:apixaban 5mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:warfarin	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:rivaroxaban 10mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 60mg:warfarin	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
placebo:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
rivaroxaban 10mg:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
rivaroxaban 20mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Indirect evidence of the safety outcome in patients with CrCl 50-80mL/min								
apixaban 2.5mg:aspirin	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:edoxaban 60mg	0	No concerns	Suspected	No concerns	Major concerns	No concerns	No concerns	Very low
apixaban 2.5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:warfarin	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:aspirin	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
apixaban 5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:edoxaban 60mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:dabigatran 150mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:edoxaban 60mg	0	No concerns	Suspected	No concerns	Major concerns	No concerns	No concerns	Very low
aspirin:placebo	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:warfarin	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
dabigatran 150mg:edoxaban 60mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:placebo	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
dabigatran 150mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
edoxaban 60mg:placebo	0	No concerns	Suspected	Some concerns	No concerns	Major concerns	No concerns	Very low
edoxaban 60mg:rivaroxaban 10mg	0	No concerns	Suspected	No concerns	Major concerns	No concerns	No concerns	Very low
edoxaban 60mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
placebo:warfarin	0	No concerns	Suspected	Some concerns	No concerns	Major concerns	No concerns	Very low
rivaroxaban 10mg:warfarin	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low

Comparison	Number of studies	Within-study bias	Across-studies bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Mixed evidence of the safety outcome in patients with CrCl >80mL/min								
apixaban 2.5mg:apixaban 5mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 2.5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:placebo	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:warfarin	1	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
aspirin:rivaroxaban 10mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
aspirin:rivaroxaban 20mg	1	No concerns	Undetected	Some concerns	No concerns	Major concerns	No concerns	Very low
dabigatran 150mg:warfarin	2	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Moderate
placebo:rivaroxaban 20mg	1	No concerns	Suspected	Some concerns	No concerns	No concerns	No concerns	Very low
rivaroxaban 10mg:rivaroxaban 20mg	1	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
rivaroxaban 20mg:warfarin	2	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Indirect evidence of the safety outcome in patients with CrCl >80mL/min								
apixaban 2.5mg:aspirin	0	No concerns	Undetected	Some concerns	No concerns	Major concerns	No concerns	Very low
apixaban 2.5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 2.5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 2.5mg:warfarin	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 5mg:aspirin	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
apixaban 5mg:dabigatran 150mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Moderate
apixaban 5mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
apixaban 5mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
aspirin:dabigatran 150mg	0	No concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
aspirin:placebo	0	No concerns	Undetected	Some concerns	No concerns	Major concerns	No concerns	Very low
aspirin:warfarin	0	No concerns	Suspected	No concerns	No concerns	Major concerns	No concerns	Very low
dabigatran 150mg:placebo	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
dabigatran 150mg:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
dabigatran 150mg:rivaroxaban 20mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
placebo:rivaroxaban 10mg	0	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
placebo:warfarin	0	No concerns	Suspected	Some concerns	No concerns	No concerns	No concerns	Very low
rivaroxaban 10mg:warfarin	0	No concerns	Suspected	Some concerns	Major concerns	No concerns	No concerns	Very low

Supplemental Item 1. Search Strategy

1) Ovid MEDLINE (1946 to 2019 Week 26) and Ovid Embase (1966-July 1, 2019)

1. exp anticoagulant agent/ or exp blood clotting inhibitor/ or exp coumarin anticoagulant/ or exp 4 hydroxycoumarin derivative/ or exp warfarin/ or exp Acenocoumarol/ or exp Dicumarol/ or exp Phenprocoumon/ or exp Phenindione/ or exp thrombin inhibitor/ or exp dabigatran/ or exp blood clotting factor 10a inhibitor/ or exp apixaban/ or exp betrixaban/ or exp edoxaban/ or exp rivaroxaban/
2. exp Renal Dialysis/ or exp Renal Replacement Therapy/ or exp Kidney Failure, Chronic/ or exp Renal Insufficiency/ or Renal Insufficiency, Chronic/ or exp Kidney Diseases/ or exp Kidney Transplantation/ or exp glomerular filtration rate/ or exp kidney function test/ or exp creatinine/ or creatinine clearance.tw.
3. exp Clinical trial/ or exp Controlled clinical trial/ or exp Randomized Controlled Trials/ or exp Random Allocation/ or exp Cohort Studies/ or exp Longitudinal Studies/ or exp Follow-Up Studies/ or exp Prospective Studies/ or exp Retrospective Studies/
4. 1 and 2 and 3
5. limit 8 to humans

2) CENTRAL (before July 2019)

#1. MeSH descriptor: [anticoagulants] explode all trees
#2. anticoagulants:ti,ab
#3. “vitamin k dependent anticoagulant*” or “vitamin k antagonist*”:ti,ab
#4. Factor Xa inhibitor$
#5. warfarin or coumadin* or aldocumar or tedicumar
#6. rivaroxaban or apixaban or edoxaban or dabigatran or betrixaban or pradaxa or eliques or eliquis or Xarelto or anticoagulation or anticoagulant$
#7. #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8. MeSH descriptor: [Kidney Diseases] explode all trees
#9. “Kidney Disease$”:ti,ab
#10. MeSH descriptor: [renal insufficiency, chronic] explode all trees
#11. 'renal insufficiency, chronic':ti,ab
#12. chronic kidney disease* or chronic renal disease* or chronic kidney insufficiency or chronic renal insufficiency or chronic kidney injury or chronic renal injury or chronic renal failure or chronic kidney failure
#13. MeSH descriptor: [dialysis] explode all trees
#14. ‘dialysis’:ti,ab
#15. ("end-stage renal" or "end-stage kidney" or "end stage renal" or "end stage kidney") next (disease or failure)
#16. MeSH descriptor: [Uremia] explode all trees
#17. ‘ur?emi*’:ti,ab

#18. MeSH descriptor: [renal replacement therapy] explode all trees

#19. 'renal replacement therapy':ti,ab

#20. h?emodialysis or h?emofiltration or h?emodiafiltration

#21. MeSH descriptor: [Peritoneal Dialysis] explode all trees

#22. 'Peritoneal Dialys*':ti,ab

#23. #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22

#24. #7 and #23